

PRJ_69

AI-Assisted Budget Allocation Optimizer for Gram Panchayats

1. Project Idea

The project is a decision support system for Gram Panchayats. It analyzes village demographic data, existing infrastructure gaps, and scheme guidelines to recommend optimal budget allocation across roads, water, sanitation, and education, along with projected impact.

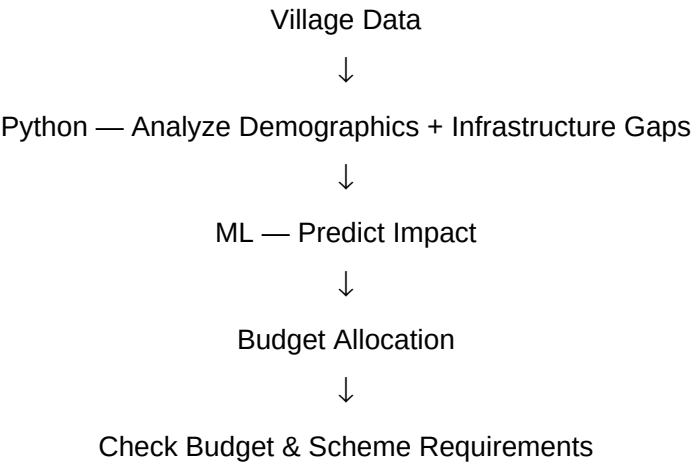
2. Main Modules

Module	Purpose	Tool
1. Data Analysis	Analyze village demographic and infrastructure data.	Python
2. Machine Learning	Predict the potential impact of different investments.	Python + ML
3. Budget Allocation	Recommend how the available budget should be distributed across sectors.	Python + ML
4. Database	Store village, demographic, infrastructure, spending and scheme-related data.	MySQL
5. Backend / API	Connect the frontend, database and Python/ML processing.	FastAPI
6. Frontend / Dashboard	Provide an interface for Panchayat officials to view and manage recommendations.	React

3. Official Technology Stack

Tool	Used For
Python	Data processing and project logic
ML	Impact prediction and intelligent recommendations
React	Frontend and dashboard
FastAPI	Backend and API
MySQL	Database

4. Overall Working



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Recommended Allocation

↓
FastAPI

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React — Panchayat Official Dashboard

MySQL stores the required village, spending and scheme-related data throughout the process.

5. Review 0 — One-Minute Explanation

“Our project is an AI-Assisted Budget Allocation Optimizer for Gram Panchayats. The main objective is to help Panchayats make data-driven decisions while allocating development funds across roads, water, sanitation and education. The system analyzes village demographic data, existing infrastructure gaps and scheme guidelines and then recommends an optimal budget allocation along with projected impact.

We have five main technologies: Python, Machine Learning, React, FastAPI and MySQL. Python and ML are used for data analysis, prediction and allocation logic. MySQL is used to store the required data. FastAPI acts as the backend API connecting the processing layer with the frontend, and React is used to build the dashboard through which Panchayat officials can view and manage the recommendations.”